

AI+ REVITALIZE & REIMAGINE EVERYTHING

STEP INTO A SMARTER TOMORROW



Foundation Models

The Watershed Moment in AI Development

Everyone's talking about AI.

It's the technology behind today's most exciting, niche-defying breakthroughs. But what makes AI so powerful? How did it evolve so fast? And how did it suddenly start generating content and understanding human languages?

This incredible story begins with the rise of **foundation models**. Unlike traditional AI, which were typically designed for one specific task, AI tools empowered by foundation models have a much broader scope. In 2017, Google introduced the **Transformer**, a revolutionary deep learning model. It later became the core technological breakthrough that supercharged the entire AI industry.



From Language and Beyond

Unlock AI's Potential in All Scenarios

To uncover the magic of AI,

let's first explore how it decodes **human language**. Imagine each word as a multi-dimensional point, with its relationship to others calculated from vast amounts of data. The introduction of the Transformer shifted AI from short-sighted to big-picture thinking in text generation.



ChatGPT

G(generative) P(pre-trained) T(Transformer)

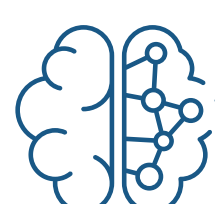
Analyzes the **entire text**.

Fills the blank based on the **overall context**

Capable of generating cohesive, nuanced responses

Multi-model AI

Models that can generate **images, videos, music**, even **interactive two-dimensional worlds**.



Artificial General Intelligence

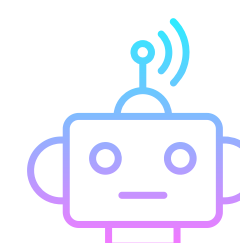
Digital human brain: capable for **any tasks**

Traditional AI

Focuses only on **nearby words**

Strings words **one by one**

Prone to awkward and irrelevant responses



The journey is still unfolding.

Like the human brain, once AI's training data surpasses a critical threshold (**10 billion+**), it gains surprising emergent abilities in a very short period.



Genie

The foundation model signals the dawn of a technological **singularity** — a tipping point for explosive growth in AI technology. An interconnected, mind-blowing future is within reach!

More Than a Chatbot

How Can AI Empower Scientists and Entrepreneurs

AI is commonly understood as conversational systems

that interact with a human in the loop. It should be noted, however, that the AI revolution in scientific research and industry has only just begun! Let's take a look at what's unfolding in **AI-powered labs and factories.**

Large Atomic Model

A Simulated World at Fingertip

The Large Atomic Model,

covering 90 elements of the periodic table, is often dubbed the **“ChatGPT for scientists.”** It is capable of **predicting** possible forms and characteristics of microscopic particles, as well as **simulating** external conditions affecting particle combinations with extensive knowledge on the properties and potential interactions of various chemical particles.

Traditionally, developing a new material involved designing the particle structure, followed by real-world testing to achieve the desired properties. With the Large Atomic Model, researchers can simulate these experiments by simply **describing the desired qualities** and **adjusting parameters** in the model.

- Reach Scale
100 to 100 million atoms
- Visuality



- Repetitive Modeling and Calculations
- Development Time
- Cost



SUCCESS CASE 1

rocket engine
flame simulation to
explore different
combustion
scenarios.



SUCCESS CASE 2

perovskite (FAPbI_3)
design with improved
photovoltaic
characteristics for
solar cell

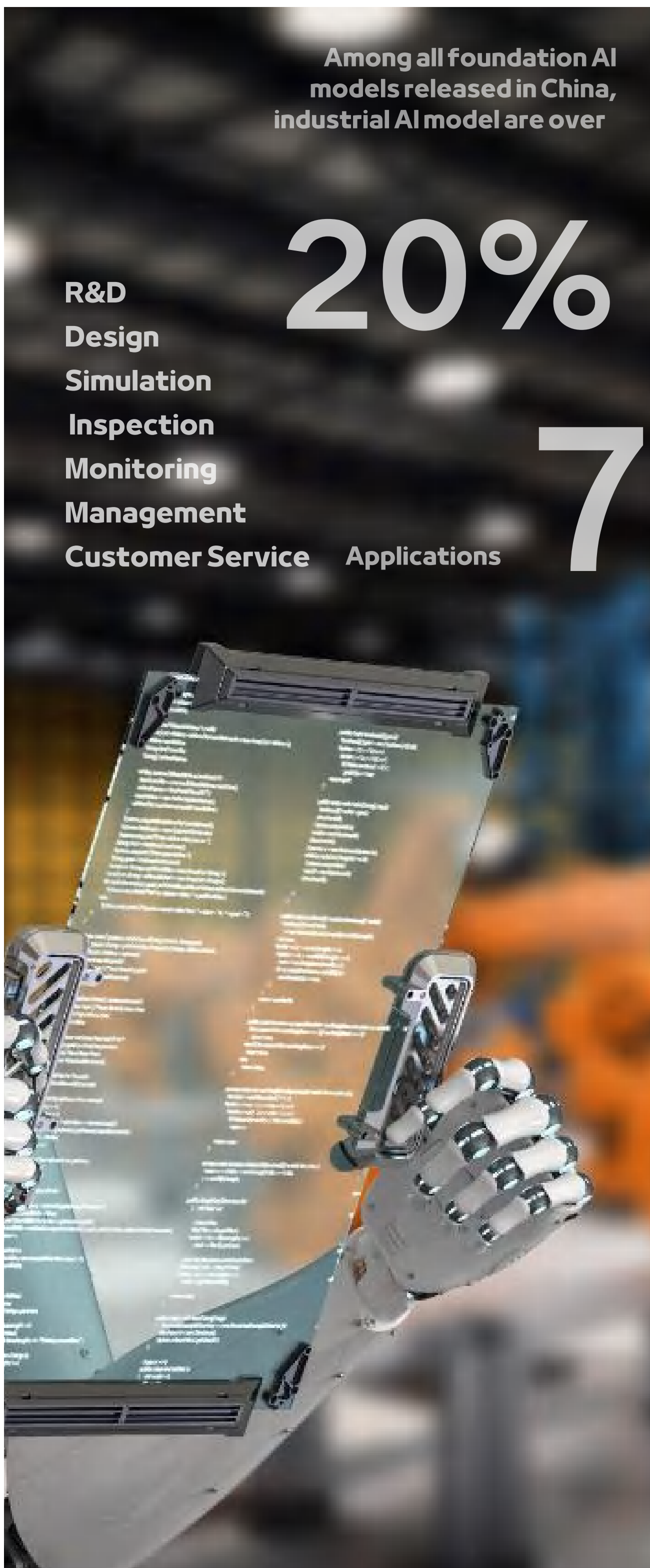


applications in
biomedicine
computational
physics
and beyond

Industrial AI Model

Amplifying Industrial Intelligence Transformation

Outside the R & D centers, AI foundation models have already started to emerge as powerful new production tools throughout the industrial processes. They're not just simulating—they get to work in physical world!



SUCCESS CASE 3: VOICED COMPONENT DESIGN

Create a cylindrical motor, with a length of 20 cm and a diameter of 16 cm. It includes a cylindrical turbine shaft and five blades.



Users describe detailed parameters of components through voice input to achieve **3D modeling**. These 3D models can then be seamlessly integrated with other design software.

Design Time down from **1 day** to **a few seconds**

Talent Training Cycle down from **1 year** to **1 month**

SUCCESS CASE 4 : AI + ROBOTS

In combination with **industrial robots**, AI can manage tasks like items selection and delivery. It can also coordinate multiple robots simultaneously, enabling them to complete more complex tasks together.

With ongoing scientific efforts to connect AI with **neural networks**, we may one day see tasks accomplished purely through thought—a powerful demonstration of “Je pense, donc je suis” (I think, therefore I am).

SUCCESS CASE 5 : MATERIAL SELECTION AND QUALITY INSPECTION

AI model helps to select materials to make integrated die-cast automotive baseplates

Intensity increased **20%**

Durability increased by **10** times

Examine X ray photos with **99.9** precision

and forecast **100+** possible flaws

Key Contributors to AI Transformation

Meet Huawei and its partners

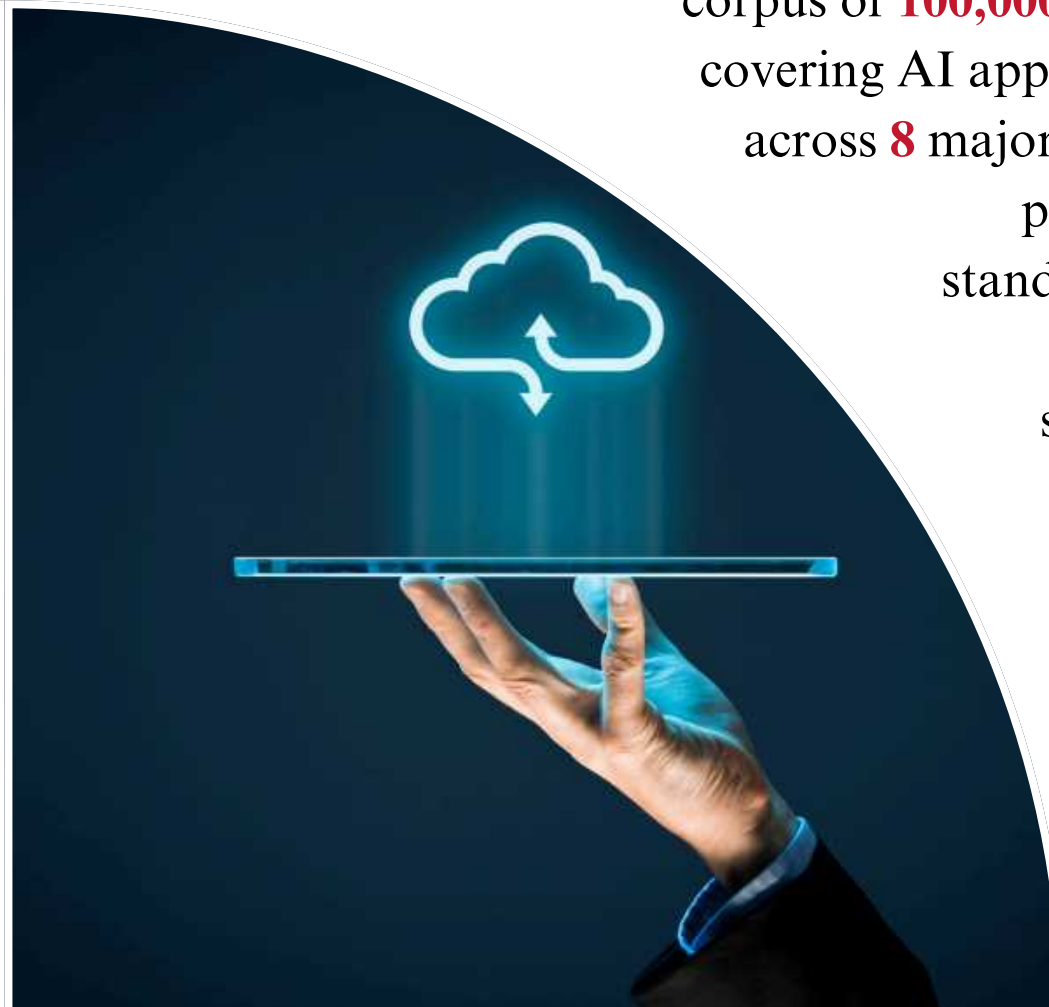
Governments

are rolling out supportive policies and enhancing data collection to stimulate AI innovation.



Scientific institutions

have created a comprehensive corpus of **100,000** entries covering AI applications across **8** major sectors, providing standardized testing services.



Huawei

Leading the way in this intelligent empowerment is Huawei. As a global ICT leader, Huawei has long been committed to launching groundbreaking technologies like **cloud computing, big data, and the Internet of Things (IoT)** to unlock AI's potential in areas including **urban management, retail, education, oil and gas**, and beyond.



*Digital and intelligent should not be a privilege for the few.
It should be a benefit for all.*

Li Peng

Corporate Senior Vice President, President of ICT Sales Service, Huawei

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the imminent era of intelligence will bring with it an enhanced quality of life for everyone. It will also usher in optimized production processes and higher operational efficiency for enterprises.

**Let's Going Intelligent
Together, Innovating the
Future!**